DATABASE MANAGEMENT SYSTEMS · ASSIGNMENT 5

Database Creation and Table Implementation

Implementing the finalized FoodExpress design with SQL DDL in MySQL

Project	FoodExpress – Online Food Delivery System	Submitted by	Dhruvan Karthik
DBMS	MySQL Community Server 8.0.46	GitHub	github.com/hidhruvan-prog
Database	`foodexpress`	Date	25 October 2026

1. Objective

Create the `foodexpress` database and implement all nine tables of the finalized design from Assignment 3 using SQL Data Definition Language (DDL), with primary keys, foreign keys and relationships defined so that the database itself enforces the links between tables. Every statement below was executed on the server; the dark panels are the actual MySQL console output.

2. Environment

Item	Value
DBMS	MySQL Community Server 8.0.46 (Windows 11, 64-bit)
Client	mysql command-line client 8.0.46
Storage engine	InnoDB (supports transactions and foreign keys)
Character set / collation	utf8mb4 / utf8mb4_0900_ai_ci (full Unicode, e.g. ₹ and regional names)
SQL mode	STRICT_TRANS_TABLES, ONLY_FULL_GROUP_BY, NO_ZERO_DATE, ERROR_FOR_DIVISION_BY_ZERO …
Script	`sql/01_create_database.sql` in the GitHub repository

3. Finalized Schema

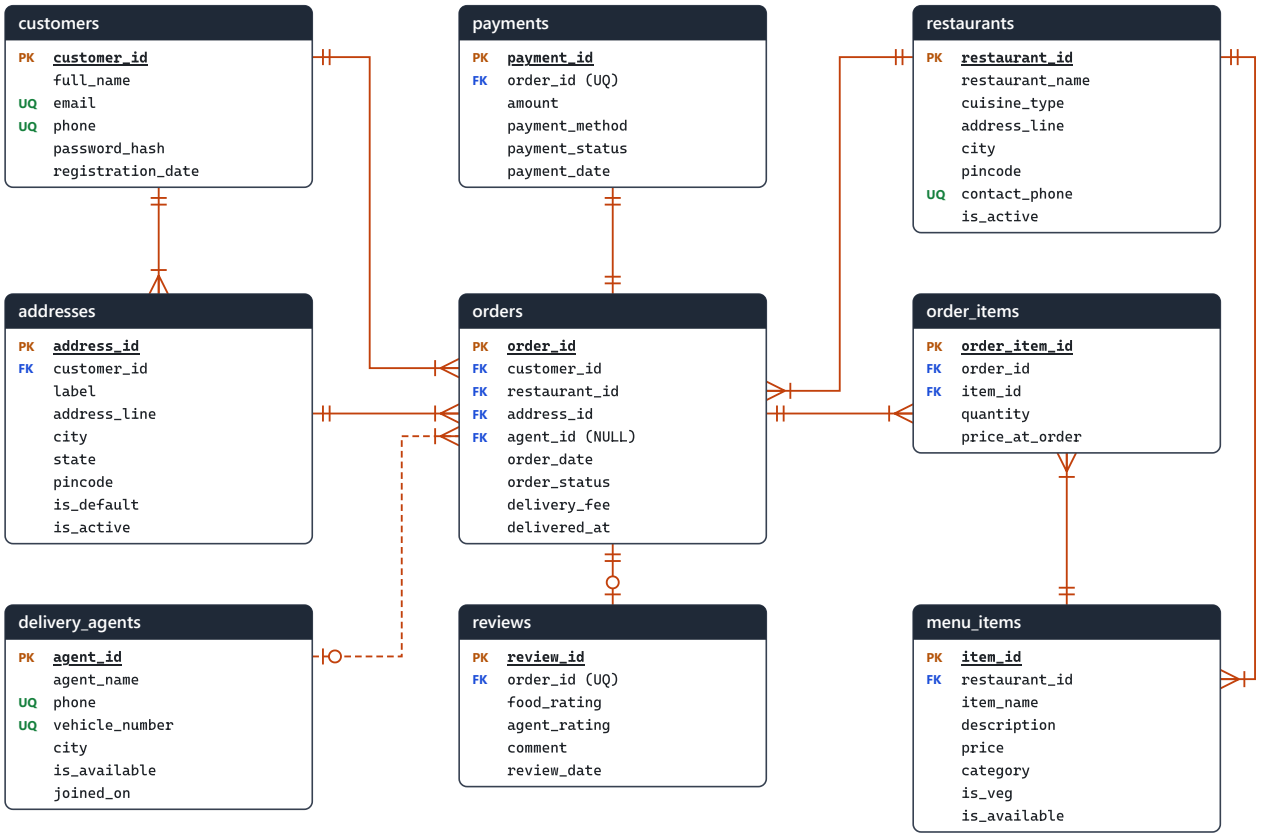


Figure 1 – Tables and relationships implemented in this assignment.

PK primary key **FK** foreign key **UQ** unique — one-to-many: bar (one) to crow's foot (many) ○ optional (zero or one) dashed: nullable foreign key

Table	Primary key	Foreign keys	Purpose
customers	customer_id	–	Registered users who place orders. Parent of addresses, orders.
addresses	address_id	customer_id → customers	Delivery addresses saved by a customer (1 : M). `UNIQUE (address_id, customer_id)` is the target of the composite foreign key in orders.
restaurants	restaurant_id	–	Partner restaurants. Parent of menu_items and orders.
menu_items	item_id	restaurant_id → restaurants	Dishes offered by each restaurant (1 : M).
delivery_agents	agent_id	–	Riders. `orders.agent_id` refers to this table and is NULL until a rider is assigned.
orders	order_id	customer_id → customers restaurant_id → restaurants (address_id, customer_id) → addresses agent_id → delivery_agents	Central transaction table linking customer, restaurant, delivery address and rider.
order_items	order_item_id	order_id → orders item_id → menu_items	Bridge table that resolves the M : N relationship between orders and menu_items; stores the price at the time of ordering.
payments	payment_id	order_id → orders	One payment per order (1 : 1; uniqueness of order_id is added in Assignment 6).
reviews	review_id	order_id → orders	Optional rating of a delivered order (1 : 1) for both the food and the rider.

4. Creating the Database

The script first drops any earlier copy so that it can be re-run, then creates the database with a Unicode character set and selects it.

```
--  FoodExpress – Online Food Delivery System
--  Assignment 5 : Database creation and table implementation (MySQL 8)
--  Finalized design (after the Assignment 3 analysis):
--    * Restaurants.address split into address_line / city / pincode
--    * stored avg_rating columns removed (derived from Reviews instead)
--    * Orders.total_amount replaced by delivery_fee + derived subtotal
--    * Orders.delivered_at added
--    * Reviews no longer repeats customer_id / restaurant_id
--      (both are reachable through order_id); agent_rating added
--    * Orders(address_id, customer_id) is a composite foreign key, so an
--      order can only use an address that belongs to the same customer
--  UNIQUE / CHECK / DEFAULT constraints are added in 02_constraints.sql

DROP DATABASE IF EXISTS foodexpress;
CREATE DATABASE foodexpress
    CHARACTER SET utf8mb4
    COLLATE utf8mb4_0900_ai_ci;
USE foodexpress;
```

mysql — foodexpress

```
mysql> DROP DATABASE IF EXISTS foodexpress;
Query OK, 11 rows affected (0.10 sec)

mysql> CREATE DATABASE foodexpress
    ->     CHARACTER SET utf8mb4
    ->     COLLATE utf8mb4_0900_ai_ci;
Query OK, 1 row affected (0.00 sec)
```

```
SHOW DATABASES LIKE 'foodexpress';
```

mysql — foodexpress

```
mysql> SHOW DATABASES LIKE 'foodexpress';
+------------------------+
| Database (foodexpress) |
+------------------------+
| foodexpress            |
+------------------------+
1 row in set (0.00 sec)
```

5. Creating the Tables

Tables are created in **dependency order**: a table can only declare a foreign key to a table that already exists. Independent "parent" tables (customers, restaurants, delivery_agents) come first, then the tables that reference them. All tables use `AUTO_INCREMENT` surrogate primary keys and name every constraint (`pk_…`, `fk_…`) so that error messages and later ALTER statements can refer to them. Referential actions are chosen per relationship (explained in Section 7).

5.1 customers

Registered users who place orders. Parent of addresses, orders.

```sql
CREATE TABLE customers (
    customer_id       INT           NOT NULL AUTO_INCREMENT,
    full_name         VARCHAR(100)  NOT NULL,
    email             VARCHAR(100)  NOT NULL,
    phone             VARCHAR(15)   NOT NULL,
    password_hash     VARCHAR(255)  NOT NULL,
    registration_date DATETIME      NOT NULL,
    CONSTRAINT pk_customers PRIMARY KEY (customer_id)
) ENGINE = InnoDB;
```

mysql — foodexpress

```
mysql> CREATE TABLE customers (
    ->     customer_id       INT           NOT NULL AUTO_INCREMENT,
    ->     full_name         VARCHAR(100)  NOT NULL,
    ->     email             VARCHAR(100)  NOT NULL,
    ->     phone             VARCHAR(15)   NOT NULL,
    ->     password_hash     VARCHAR(255)  NOT NULL,
    ->     registration_date DATETIME      NOT NULL,
    ->     CONSTRAINT pk_customers PRIMARY KEY (customer_id)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)
```

5.2 addresses

Delivery addresses saved by a customer (1 : M). `UNIQUE (address_id, customer_id)` is the target of the composite foreign key in orders.

```sql
CREATE TABLE addresses (
    address_id    INT           NOT NULL AUTO_INCREMENT,
    customer_id   INT           NOT NULL,
    label         VARCHAR(20)   NOT NULL,          -- Home / Work / Other
    address_line  VARCHAR(255)  NOT NULL,
    city          VARCHAR(50)   NOT NULL,
    state         VARCHAR(50)   NOT NULL,
    pincode       CHAR(6)       NOT NULL,
    is_default    BOOLEAN       NOT NULL,
    is_active     BOOLEAN       NOT NULL,
    CONSTRAINT pk_addresses PRIMARY KEY (address_id),
    -- candidate key used as the target of the composite FK in orders
    CONSTRAINT uq_addresses_id_customer UNIQUE (address_id, customer_id),
    CONSTRAINT fk_addresses_customer
        FOREIGN KEY (customer_id) REFERENCES customers (customer_id)
        ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE = InnoDB;
```

mysql — foodexpress

```
mysql> CREATE TABLE addresses (
    ->     address_id    INT           NOT NULL AUTO_INCREMENT,
    ->     customer_id   INT           NOT NULL,
    ->     label         VARCHAR(20)   NOT NULL,
    ->     address_line  VARCHAR(255)  NOT NULL,
    ->     city          VARCHAR(50)   NOT NULL,
    ->     state         VARCHAR(50)   NOT NULL,
    ->     pincode       CHAR(6)       NOT NULL,
    ->     is_default    BOOLEAN       NOT NULL,
    ->     is_active     BOOLEAN       NOT NULL,
    ->     CONSTRAINT pk_addresses PRIMARY KEY (address_id),
    ->
    ->     CONSTRAINT uq_addresses_id_customer UNIQUE (address_id, customer_id),
    ->     CONSTRAINT fk_addresses_customer
    ->         FOREIGN KEY (customer_id) REFERENCES customers (customer_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.05 sec)
```

5.3 restaurants

Partner restaurants. Parent of menu_items and orders.

```sql
CREATE TABLE restaurants (
    restaurant_id    INT           NOT NULL AUTO_INCREMENT,
    restaurant_name  VARCHAR(100)  NOT NULL,
    cuisine_type     VARCHAR(50)   NOT NULL,
    address_line     VARCHAR(255)  NOT NULL,
    city             VARCHAR(50)   NOT NULL,
    pincode          CHAR(6)       NOT NULL,
    contact_phone    VARCHAR(15)   NOT NULL,
    is_active        BOOLEAN       NOT NULL,
    CONSTRAINT pk_restaurants PRIMARY KEY (restaurant_id)
) ENGINE = InnoDB;
```

mysql — foodexpress

```
mysql> CREATE TABLE restaurants (
    ->     restaurant_id    INT           NOT NULL AUTO_INCREMENT,
    ->     restaurant_name  VARCHAR(100)  NOT NULL,
    ->     cuisine_type     VARCHAR(50)   NOT NULL,
    ->     address_line     VARCHAR(255)  NOT NULL,
    ->     city             VARCHAR(50)   NOT NULL,
    ->     pincode          CHAR(6)       NOT NULL,
    ->     contact_phone    VARCHAR(15)   NOT NULL,
    ->     is_active        BOOLEAN       NOT NULL,
    ->     CONSTRAINT pk_restaurants PRIMARY KEY (restaurant_id)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)
```

5.4 menu_items

Dishes offered by each restaurant (1 : M).

```sql
CREATE TABLE menu_items (
    item_id        INT           NOT NULL AUTO_INCREMENT,
    restaurant_id  INT           NOT NULL,
    item_name      VARCHAR(100)  NOT NULL,
    description    VARCHAR(255)  NULL,
    price          DECIMAL(8,2)  NOT NULL,
    category       VARCHAR(50)   NOT NULL,
    is_veg         BOOLEAN       NOT NULL,
    is_available   BOOLEAN       NOT NULL,
    CONSTRAINT pk_menu_items PRIMARY KEY (item_id),
    CONSTRAINT fk_menu_items_restaurant
        FOREIGN KEY (restaurant_id) REFERENCES restaurants (restaurant_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;
```

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```
mysql> CREATE TABLE menu_items (
    ->     item_id        INT           NOT NULL AUTO_INCREMENT,
    ->     restaurant_id  INT           NOT NULL,
    ->     item_name      VARCHAR(100)  NOT NULL,
    ->     description    VARCHAR(255)  NULL,
    ->     price          DECIMAL(8,2)  NOT NULL,
    ->     category       VARCHAR(50)   NOT NULL,
    ->     is_veg         BOOLEAN       NOT NULL,
    ->     is_available   BOOLEAN       NOT NULL,
    ->     CONSTRAINT pk_menu_items PRIMARY KEY (item_id),
    ->     CONSTRAINT fk_menu_items_restaurant
    ->         FOREIGN KEY (restaurant_id) REFERENCES restaurants (restaurant_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.07 sec)
```

5.5 delivery_agents

Riders. `orders.agent_id` refers to this table and is NULL until a rider is assigned.

```sql
CREATE TABLE delivery_agents (
    agent_id        INT           NOT NULL AUTO_INCREMENT,
    agent_name      VARCHAR(100)  NOT NULL,
    phone           VARCHAR(15)   NOT NULL,
    vehicle_number  VARCHAR(20)   NOT NULL,
    city            VARCHAR(50)   NOT NULL,
    is_available    BOOLEAN       NOT NULL,
    joined_on       DATE          NOT NULL,
    CONSTRAINT pk_delivery_agents PRIMARY KEY (agent_id)
) ENGINE = InnoDB;
```

mysql — foodexpress

```
mysql> CREATE TABLE delivery_agents (
    ->     agent_id        INT           NOT NULL AUTO_INCREMENT,
    ->     agent_name      VARCHAR(100)  NOT NULL,
    ->     phone           VARCHAR(15)   NOT NULL,
    ->     vehicle_number  VARCHAR(20)   NOT NULL,
    ->     city            VARCHAR(50)   NOT NULL,
    ->     is_available    BOOLEAN       NOT NULL,
    ->     joined_on       DATE          NOT NULL,
    ->     CONSTRAINT pk_delivery_agents PRIMARY KEY (agent_id)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)
```

5.6 orders

Central transaction table linking customer, restaurant, delivery address and rider.

```
CREATE TABLE orders (
    order_id       INT            NOT NULL AUTO_INCREMENT,
    customer_id    INT            NOT NULL,
    restaurant_id  INT            NOT NULL,
    address_id     INT            NOT NULL,
    agent_id       INT            NULL,           -- NULL until assigned
    order_date     DATETIME       NOT NULL,
    order_status   ENUM('Placed','Preparing','Out for Delivery',
                        'Delivered','Cancelled') NOT NULL,
    delivery_fee   DECIMAL(6,2)   NOT NULL,
    delivered_at   DATETIME       NULL,
    CONSTRAINT pk_orders PRIMARY KEY (order_id),
    CONSTRAINT fk_orders_customer
        FOREIGN KEY (customer_id) REFERENCES customers (customer_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_orders_restaurant
        FOREIGN KEY (restaurant_id) REFERENCES restaurants (restaurant_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    -- the delivery address must belong to the customer placing the order
    CONSTRAINT fk_orders_address_owner
        FOREIGN KEY (address_id, customer_id)
        REFERENCES addresses (address_id, customer_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_orders_agent
        FOREIGN KEY (agent_id) REFERENCES delivery_agents (agent_id)
        ON UPDATE CASCADE ON DELETE SET NULL
) ENGINE = InnoDB;
```

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```
mysql> CREATE TABLE orders (
    ->     order_id       INT            NOT NULL AUTO_INCREMENT,
    ->     customer_id    INT            NOT NULL,
    ->     restaurant_id  INT            NOT NULL,
    ->     address_id     INT            NOT NULL,
    ->     agent_id       INT            NULL,
    ->     order_date     DATETIME       NOT NULL,
    ->     order_status   ENUM('Placed','Preparing','Out for Delivery',
    ->                         'Delivered','Cancelled') NOT NULL,
    ->     delivery_fee   DECIMAL(6,2)   NOT NULL,
    ->     delivered_at   DATETIME       NULL,
    ->     CONSTRAINT pk_orders PRIMARY KEY (order_id),
    ->     CONSTRAINT fk_orders_customer
    ->         FOREIGN KEY (customer_id) REFERENCES customers (customer_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->     CONSTRAINT fk_orders_restaurant
    ->         FOREIGN KEY (restaurant_id) REFERENCES restaurants (restaurant_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->
    ->     CONSTRAINT fk_orders_address_owner
    ->         FOREIGN KEY (address_id, customer_id)
    ->         REFERENCES addresses (address_id, customer_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->     CONSTRAINT fk_orders_agent
    ->         FOREIGN KEY (agent_id) REFERENCES delivery_agents (agent_id)
    ->         ON UPDATE CASCADE ON DELETE SET NULL
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.07 sec)
```

5.7 order_items

Bridge table that resolves the M : N relationship between orders and menu_items; stores the price at the time of ordering.

```
CREATE TABLE order_items (
    order_item_id   INT           NOT NULL AUTO_INCREMENT,
    order_id        INT           NOT NULL,
    item_id         INT           NOT NULL,
    quantity        INT           NOT NULL,
    price_at_order  DECIMAL(8,2)  NOT NULL,       -- price snapshot
    CONSTRAINT pk_order_items PRIMARY KEY (order_item_id),
    CONSTRAINT fk_order_items_order
        FOREIGN KEY (order_id) REFERENCES orders (order_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_order_items_item
        FOREIGN KEY (item_id) REFERENCES menu_items (item_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;
```

mysql — foodexpress

```
mysql> CREATE TABLE order_items (
    ->     order_item_id   INT           NOT NULL AUTO_INCREMENT,
    ->     order_id        INT           NOT NULL,
    ->     item_id         INT           NOT NULL,
    ->     quantity        INT           NOT NULL,
    ->     price_at_order  DECIMAL(8,2)  NOT NULL,
    ->     CONSTRAINT pk_order_items PRIMARY KEY (order_item_id),
    ->     CONSTRAINT fk_order_items_order
    ->         FOREIGN KEY (order_id) REFERENCES orders (order_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_order_items_item
    ->         FOREIGN KEY (item_id) REFERENCES menu_items (item_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.06 sec)
```

5.8 payments

One payment per order (1 : 1; uniqueness of order_id is added in Assignment 6).

```
CREATE TABLE payments (
    payment_id      INT           NOT NULL AUTO_INCREMENT,
    order_id        INT           NOT NULL,
    amount          DECIMAL(10,2)  NOT NULL,
    payment_method  ENUM('Cash','Card','UPI','Wallet') NOT NULL,
    payment_status  ENUM('Pending','Completed','Failed','Refunded') NOT NULL,
    payment_date    DATETIME      NOT NULL,
    CONSTRAINT pk_payments PRIMARY KEY (payment_id),
    CONSTRAINT fk_payments_order
        FOREIGN KEY (order_id) REFERENCES orders (order_id)
        ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE = InnoDB;
```

mysql — foodexpress

```
mysql> CREATE TABLE payments (
    ->     payment_id      INT            NOT NULL AUTO_INCREMENT,
    ->     order_id        INT            NOT NULL,
    ->     amount          DECIMAL(10,2)  NOT NULL,
    ->     payment_method  ENUM('Cash','Card','UPI','Wallet') NOT NULL,
    ->     payment_status  ENUM('Pending','Completed','Failed','Refunded') NOT NULL,
    ->     payment_date    DATETIME       NOT NULL,
    ->     CONSTRAINT pk_payments PRIMARY KEY (payment_id),
    ->     CONSTRAINT fk_payments_order
    ->         FOREIGN KEY (order_id) REFERENCES orders (order_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.06 sec)
```

5.9 reviews

Optional rating of a delivered order (1 : 1) for both the food and the rider.

```
CREATE TABLE reviews (
    review_id      INT           NOT NULL AUTO_INCREMENT,
    order_id       INT           NOT NULL,
    food_rating    TINYINT       NOT NULL,
    agent_rating   TINYINT       NULL,
    comment        VARCHAR(500)  NULL,
    review_date    DATETIME      NOT NULL,
    CONSTRAINT pk_reviews PRIMARY KEY (review_id),
    CONSTRAINT fk_reviews_order
        FOREIGN KEY (order_id) REFERENCES orders (order_id)
        ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE = InnoDB;
```

mysql — foodexpress

```
mysql> CREATE TABLE reviews (
    ->     review_id      INT           NOT NULL AUTO_INCREMENT,
    ->     order_id       INT           NOT NULL,
    ->     food_rating    TINYINT       NOT NULL,
    ->     agent_rating   TINYINT       NULL,
    ->     comment        VARCHAR(500)  NULL,
    ->     review_date    DATETIME      NOT NULL,
    ->     CONSTRAINT pk_reviews PRIMARY KEY (review_id),
    ->     CONSTRAINT fk_reviews_order
    ->         FOREIGN KEY (order_id) REFERENCES orders (order_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.05 sec)
```

6. Views for Derived Data

As decided in Assignment 3, order totals and restaurant ratings are not stored in tables (that would be redundant). Two views compute them on demand from the base tables.

```
CREATE VIEW v_order_totals AS
SELECT o.order_id,
       SUM(oi.quantity * oi.price_at_order)                 AS subtotal,
       o.delivery_fee,
       SUM(oi.quantity * oi.price_at_order) + o.delivery_fee AS total_amount
FROM orders o
JOIN order_items oi ON oi.order_id = o.order_id
GROUP BY o.order_id, o.delivery_fee;
```

mysql — foodexpress

```
mysql> CREATE VIEW v_order_totals AS
    -> SELECT o.order_id,
    ->        SUM(oi.quantity * oi.price_at_order)                 AS subtotal,
    ->        o.delivery_fee,
    ->        SUM(oi.quantity * oi.price_at_order) + o.delivery_fee AS total_amount
    -> FROM orders o
    -> JOIN order_items oi ON oi.order_id = o.order_id
    -> GROUP BY o.order_id, o.delivery_fee;
Query OK, 0 rows affected (0.01 sec)
```

```sql
CREATE VIEW v_restaurant_ratings AS
SELECT r.restaurant_id,
       r.restaurant_name,
       COUNT(rv.review_id)          AS review_count,
       ROUND(AVG(rv.food_rating),1) AS avg_rating
FROM restaurants r
LEFT JOIN orders  o  ON o.restaurant_id = r.restaurant_id
LEFT JOIN reviews rv ON rv.order_id     = o.order_id
GROUP BY r.restaurant_id, r.restaurant_name;
```

mysql — foodexpress

```
mysql> CREATE VIEW v_restaurant_ratings AS
    -> SELECT r.restaurant_id,
    ->        r.restaurant_name,
    ->        COUNT(rv.review_id)          AS review_count,
    ->        ROUND(AVG(rv.food_rating),1) AS avg_rating
    -> FROM restaurants r
    -> LEFT JOIN orders  o  ON o.restaurant_id = r.restaurant_id
    -> LEFT JOIN reviews rv ON rv.order_id     = o.order_id
    -> GROUP BY r.restaurant_id, r.restaurant_name;
Query OK, 0 rows affected (0.01 sec)
```

7. Verifying the Tables and Relationships

7.1 Tables created

```sql
SHOW FULL TABLES;
```

mysql — foodexpress

```
mysql> SHOW FULL TABLES;
+----------------------+------------+
| Tables_in_foodexpress | Table_type |
+----------------------+------------+
| addresses            | BASE TABLE |
| customers            | BASE TABLE |
| delivery_agents      | BASE TABLE |
| menu_items           | BASE TABLE |
| order_items          | BASE TABLE |
| orders               | BASE TABLE |
| payments             | BASE TABLE |
| restaurants          | BASE TABLE |
| reviews              | BASE TABLE |
| v_order_totals       | VIEW       |
| v_restaurant_ratings | VIEW       |
+----------------------+------------+
11 rows in set (0.00 sec)
```

7.2 Table structures

`DESCRIBE` confirms each column's data type, nullability and key. `PRI` marks the primary key, `MUL` an indexed foreign-key column and `UNI` a unique column. Defaults and CHECK constraints are added in Assignment 6, so the *Default* column is still NULL here.

```
DESCRIBE customers;
```

mysql — foodexpress

```
mysql> DESCRIBE customers;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| customer_id       | int          | NO   | PRI | NULL    | auto_increment |
| full_name         | varchar(100) | NO   |     | NULL    |                |
| email             | varchar(100) | NO   |     | NULL    |                |
| phone             | varchar(15)  | NO   |     | NULL    |                |
| password_hash     | varchar(255) | NO   |     | NULL    |                |
| registration_date | datetime     | NO   |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

```
DESCRIBE addresses;
```

mysql — foodexpress

```
mysql> DESCRIBE addresses;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| address_id   | int          | NO   | PRI | NULL    | auto_increment |
| customer_id  | int          | NO   | MUL | NULL    |                |
| label        | varchar(20)  | NO   |     | NULL    |                |
| address_line | varchar(255) | NO   |     | NULL    |                |
| city         | varchar(50)  | NO   |     | NULL    |                |
| state        | varchar(50)  | NO   |     | NULL    |                |
| pincode      | char(6)      | NO   |     | NULL    |                |
| is_default   | tinyint(1)   | NO   |     | NULL    |                |
| is_active    | tinyint(1)   | NO   |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

```
DESCRIBE restaurants;
```

mysql — foodexpress

```
mysql> DESCRIBE restaurants;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| restaurant_id   | int          | NO   | PRI | NULL    | auto_increment |
| restaurant_name | varchar(100) | NO   |     | NULL    |                |
| cuisine_type    | varchar(50)  | NO   |     | NULL    |                |
| address_line    | varchar(255) | NO   |     | NULL    |                |
| city            | varchar(50)  | NO   |     | NULL    |                |
| pincode         | char(6)      | NO   |     | NULL    |                |
| contact_phone   | varchar(15)  | NO   |     | NULL    |                |
| is_active       | tinyint(1)   | NO   |     | NULL    |                |
+-----------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

```
DESCRIBE menu_items;
```

mysql — foodexpress

```
mysql> DESCRIBE menu_items;
+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| item_id       | int          | NO   | PRI | NULL    | auto_increment |
| restaurant_id | int          | NO   | MUL | NULL    |                |
| item_name     | varchar(100) | NO   |     | NULL    |                |
| description   | varchar(255) | YES  |     | NULL    |                |
| price         | decimal(8,2) | NO   |     | NULL    |                |
| category      | varchar(50)  | NO   |     | NULL    |                |
| is_veg        | tinyint(1)   | NO   |     | NULL    |                |
| is_available  | tinyint(1)   | NO   |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
8 rows in set (0.01 sec)
```

```
DESCRIBE delivery_agents;
```

mysql — foodexpress

```
mysql> DESCRIBE delivery_agents;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| agent_id       | int          | NO   | PRI | NULL    | auto_increment |
| agent_name     | varchar(100) | NO   |     | NULL    |                |
| phone          | varchar(15)  | NO   |     | NULL    |                |
| vehicle_number | varchar(20)  | NO   |     | NULL    |                |
| city           | varchar(50)  | NO   |     | NULL    |                |
| is_available   | tinyint(1)   | NO   |     | NULL    |                |
| joined_on      | date         | NO   |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

```
DESCRIBE orders;
```

mysql — foodexpress

```
mysql> DESCRIBE orders;
+---------------+-----------------------------------------------------------------------+------+-----+---------+----------------+
| Field         | Type                                                                  | Null | Key | Default | Extra          |
+---------------+-----------------------------------------------------------------------+------+-----+---------+----------------+
| order_id      | int                                                                   | NO   | PRI | NULL    | auto_increment |
| customer_id   | int                                                                   | NO   | MUL | NULL    |                |
| restaurant_id | int                                                                   | NO   | MUL | NULL    |                |
| address_id    | int                                                                   | NO   | MUL | NULL    |                |
| agent_id      | int                                                                   | YES  | MUL | NULL    |                |
| order_date    | datetime                                                              | NO   |     | NULL    |                |
| order_status  | enum('Placed','Preparing','Out for Delivery','Delivered','Cancelled') | NO   |     | NULL    |                |
| delivery_fee  | decimal(6,2)                                                          | NO   |     | NULL    |                |
| delivered_at  | datetime                                                              | YES  |     | NULL    |                |
+---------------+-----------------------------------------------------------------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

```
DESCRIBE order_items;
```

mysql — foodexpress

```
mysql> DESCRIBE order_items;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| order_item_id  | int          | NO   | PRI | NULL    | auto_increment |
| order_id       | int          | NO   | MUL | NULL    |                |
| item_id        | int          | NO   | MUL | NULL    |                |
| quantity       | int          | NO   |     | NULL    |                |
| price_at_order | decimal(8,2) | NO   |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
5 rows in set (0.01 sec)
```

```
DESCRIBE payments;
```

mysql — foodexpress

```
mysql> DESCRIBE payments;
+----------------+-----------------------------------------------------+------+-----+---------+----------------+
| Field          | Type                                                | Null | Key | Default | Extra          |
+----------------+-----------------------------------------------------+------+-----+---------+----------------+
| payment_id     | int                                                 | NO   | PRI | NULL    | auto_increment |
| order_id       | int                                                 | NO   | MUL | NULL    |                |
| amount         | decimal(10,2)                                       | NO   |     | NULL    |                |
| payment_method | enum('Cash','Card','UPI','Wallet')                  | NO   |     | NULL    |                |
| payment_status | enum('Pending','Completed','Failed','Refunded')     | NO   |     | NULL    |                |
| payment_date   | datetime                                            | NO   |     | NULL    |                |
+----------------+-----------------------------------------------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

```
DESCRIBE reviews;
```

mysql — foodexpress

```
mysql> DESCRIBE reviews;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| review_id    | int          | NO   | PRI | NULL    | auto_increment |
| order_id     | int          | NO   | MUL | NULL    |                |
| food_rating  | tinyint      | NO   |     | NULL    |                |
| agent_rating | tinyint      | YES  |     | NULL    |                |
| comment      | varchar(500) | YES  |     | NULL    |                |
| review_date  | datetime     | NO   |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

7.3 Foreign-key relationships recorded in the data dictionary

The query below reads MySQL's `information_schema` to list every foreign key with the column it uses, the table and column it references, and its ON UPDATE / ON DELETE rule.

```
SELECT k.TABLE_NAME            AS child_table,
       k.COLUMN_NAME           AS fk_column,
       k.REFERENCED_TABLE_NAME AS parent_table,
       k.REFERENCED_COLUMN_NAME AS parent_column,
       r.UPDATE_RULE, r.DELETE_RULE,
       k.CONSTRAINT_NAME
FROM information_schema.KEY_COLUMN_USAGE k
JOIN information_schema.REFERENTIAL_CONSTRAINTS r
  ON r.CONSTRAINT_SCHEMA = k.CONSTRAINT_SCHEMA
 AND r.CONSTRAINT_NAME   = k.CONSTRAINT_NAME
WHERE k.TABLE_SCHEMA = 'foodexpress'
ORDER BY k.TABLE_NAME, k.CONSTRAINT_NAME, k.ORDINAL_POSITION;
```

mysql — foodexpress

```
mysql> SELECT k.TABLE_NAME            AS child_table,
    ->        k.COLUMN_NAME           AS fk_column,
    ->        k.REFERENCED_TABLE_NAME AS parent_table,
    ->        k.REFERENCED_COLUMN_NAME AS parent_column,
    ->        r.UPDATE_RULE, r.DELETE_RULE,
    ->        k.CONSTRAINT_NAME
    -> FROM information_schema.KEY_COLUMN_USAGE k
    -> JOIN information_schema.REFERENTIAL_CONSTRAINTS r
    ->   ON r.CONSTRAINT_SCHEMA = k.CONSTRAINT_SCHEMA
    ->  AND r.CONSTRAINT_NAME   = k.CONSTRAINT_NAME
    -> WHERE k.TABLE_SCHEMA = 'foodexpress'
    -> ORDER BY k.TABLE_NAME, k.CONSTRAINT_NAME, k.ORDINAL_POSITION;
+-------------+--------------+-----------------+--------------+-------------+-------------+--------------------------+
| child_table | fk_column    | parent_table    | parent_column | UPDATE_RULE | DELETE_RULE | CONSTRAINT_NAME          |
+-------------+--------------+-----------------+--------------+-------------+-------------+--------------------------+
| addresses   | customer_id  | customers       | customer_id  | CASCADE     | CASCADE     | fk_addresses_customer    |
| menu_items  | restaurant_id | restaurants     | restaurant_id | CASCADE     | RESTRICT    | fk_menu_items_restaurant |
| order_items | item_id      | menu_items      | item_id      | CASCADE     | RESTRICT    | fk_order_items_item      |
| order_items | order_id     | orders          | order_id     | CASCADE     | CASCADE     | fk_order_items_order     |
| orders      | address_id   | addresses       | address_id   | CASCADE     | RESTRICT    | fk_orders_address_owner  |
| orders      | customer_id  | addresses       | customer_id  | CASCADE     | RESTRICT    | fk_orders_address_owner  |
| orders      | agent_id     | delivery_agents | agent_id     | CASCADE     | SET NULL    | fk_orders_agent          |
| orders      | customer_id  | customers       | customer_id  | CASCADE     | RESTRICT    | fk_orders_customer       |
| orders      | restaurant_id | restaurants     | restaurant_id | CASCADE     | RESTRICT    | fk_orders_restaurant     |
| payments    | order_id     | orders          | order_id     | CASCADE     | CASCADE     | fk_payments_order        |
| reviews     | order_id     | orders          | order_id     | CASCADE     | CASCADE     | fk_reviews_order         |
+-------------+--------------+-----------------+--------------+-------------+-------------+--------------------------+
11 rows in set (0.00 sec)
```

7.4 Relationship summary

Relationship	Cardinality	ON DELETE	Reason
customers → addresses	1 : M	CASCADE	A customer's saved addresses have no meaning without the customer.
customers → orders	1 : M	RESTRICT	Order history must never disappear because an account is removed.
addresses → orders (composite)	1 : M	RESTRICT	Past orders keep their delivery address; also guarantees the address belongs to the same customer.
restaurants → menu_items	1 : M	RESTRICT	A restaurant with a menu is deactivated (is_active), not deleted.
restaurants → orders	1 : M	RESTRICT	Protects order history.
delivery_agents → orders	1 : M (optional)	SET NULL	If a rider record is removed, the order remains, just without a rider.
orders → order_items	1 : M	CASCADE	Line items are part of the order.
menu_items → order_items	1 : M	RESTRICT	A dish that appears in past orders cannot be deleted (mark unavailable instead).
orders → payments	1 : 1	CASCADE	A payment record belongs to exactly one order.
orders → reviews	1 : 0..1	CASCADE	A review belongs to exactly one order.

All foreign keys use `ON UPDATE CASCADE`, so if a key value is ever changed the change flows to the child rows automatically. The many-to-many relationship between orders and menu items is implemented through `order_items`, which holds one row per (order, item) pair.

8. Conclusion

The `foodexpress` database was created with nine InnoDB tables and two views that implement the finalized design exactly. Every table has a primary key, ten foreign-key constraints connect the tables with referential actions chosen for each relationship, and a composite foreign key ensures that an order is delivered to an address owned by the ordering customer. The data dictionary queries confirm that the tables, columns and relationships exist as designed. UNIQUE, CHECK and DEFAULT constraints are added in Assignment 6.